1. **Simple Interest** Suppose Liz borrows \$1000 and agrees to pay it back in 3 years with 5% simple interest added each year.

   (a) How much **interest** will she pay? In addition to your answer, write down your computation, including the decimal values you would plug into a calculator.

   Recall $I = (Pr)t = \boxed{(1000)(0.05)(3)} = \$150$

   P, t, r

   \$ added in 1 year

   5% as decimal

   computation

   answer

   (b) How much will she owe **in total** at the end of three years? Write down the computation and the answer.

   A

   Recall $A = P + I = 1000 + 150 = \$1150$

   computation

   answer

2. **Annual Interest Rate (APR)** Annual interest rates are pro-rated by time.

   (a) Definition: An APR of r% paid n times year means

   n times each year $\frac{r\%}{n}$ is added.

   (b) So an APR of 5% paid quarterly means

   quarterly means 4x per year. $\frac{5\%}{4} = \frac{0.05}{4} = 0.0125$

   So 4 times each year 1.25% interest is added.

3. Determine n for each word below.

word:	annually	semi-annually	quarterly	monthly	weekly	daily
n times per year	1	2	4	12	52	365

rough approximations

Compound Interest

4. What is **compound** interest and how is it different from **Simple** Interest?

   Compound interest is a percent of P+I, where I is whatever interest has accumulated over time.

   Simple interest is a percent of P only

Note to Instructors: Explaining this part – especially why F3 not C3 is the most important part.

5. Suppose \$1000 will be invested in an account that accrues **compound** interest at an APR of 3% compounded monthly. We will set up a spreadsheet to model this account assuming no other deposits or withdrawals.

   (a) Open a new spreadsheet and type column headers in cells `A1, B1, C1, D1, E1, F1`. (See below.)

   (b) In row 2, type the variables for reference. (hint: `A2` is r.)

   (c) In row 3, enter the **start** values for this specific problem. Assume time t is in years.

   (d) In rows 4 and 5, enter the values/formulas and the end of 1 month and 2 months.

   (e) Use the pull down feature to determine the values in the account at the end of 2 years.

	A	B	C	D	E	F
1	`Interest Rate`	`Times per Year`	`Principal`	`Time Elapsed`	`Interest`	`Total`
2	r	n	P	t	I	A
3	$=0.03$	$=12$	$=1000$	$=0$	$=0$	$=1000$
4	$=0.03$	$=12$	$=1000$	$=D3+1/B4$	$=$	$=F3+E4$
5	$=0.03$	$=12$	$=1000$	$=D4+1/B5$	$=$	$=F4+E5$

start

$$\boxed{=(F3)*(A4/B4)}$$

$$=(F4)(A5/B5)$$

6. Write out the month-by-month computations and look for a pattern.

1 month: $A = P + P\left(\frac{r}{n}\right) = P\left(1+\frac{r}{n}\right)$

2 months: $A = P\left(1+\frac{r}{n}\right) + P\left(1+\frac{r}{n}\right)\left(\frac{r}{n}\right) = P\left(1+\frac{r}{n}\right)\left(1+\frac{r}{n}\right) = P\left(1+\frac{r}{n}\right)^2$

3 months: $A = P\left(1+\frac{r}{n}\right)^2 + P\left(1+\frac{r}{n}\right)^2\left(\frac{r}{n}\right) = P\left(1+\frac{r}{n}\right)^2\left(1+\frac{r}{n}\right) = P\left(1+\frac{r}{n}\right)^3$

$\vdots$

K months: $A = P\left(1+\frac{r}{n}\right)^k$ Wait! time $\boxed{t}$ is in YEARS! $A = P\left(1+\frac{r}{n}\right)^{tn}$

7. With a principal of P dollars earning r% interest compounded n times per year for t years will result in a total of A dollars where

$$A = P\left(1+\frac{r}{n}\right)^{tn}$$